

Final Report
**Fundus Glaucoma Detection using Convolutional Neural Network
(CNN)**

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I. Introduction

Fundus Glaucoma Overview :

A. Disease Fundamental

Glaucoma is a group of eye conditions characterized by damage to the optic nerve, which can lead to vision loss or blindness if left untreated. It is typically associated with elevated intraocular pressure (IOP) and can be categorized into several types, including primary open-angle glaucoma (POAG), angle-closure glaucoma, and normal-tension glaucoma, among others. The term "fundus" refers to the back part of the eye, which includes the retina, optic nerve head, and blood vessels. An examination of the fundus is often part of the diagnostic process for various eye conditions, including glaucoma.

Glaucoma is caused by the increased IOP in eyes and is hence considered as among one of the reasons for vision misfortune that in the long run prompts visual impairment (Bokhari, Sharif, Yasmin, & Fernandes [2017]. (<https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/full/10.1002/jemt.23094#jemt23094-bib-0025>))

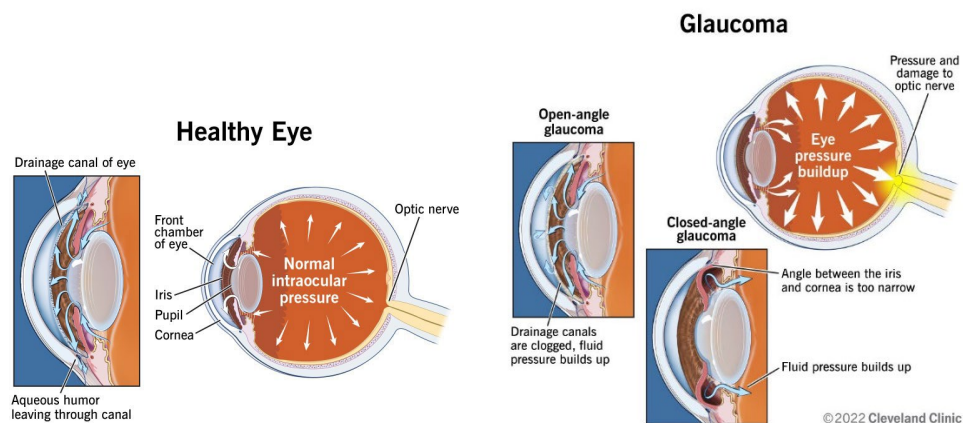


Figure 1. Fundus Glaucoma Images

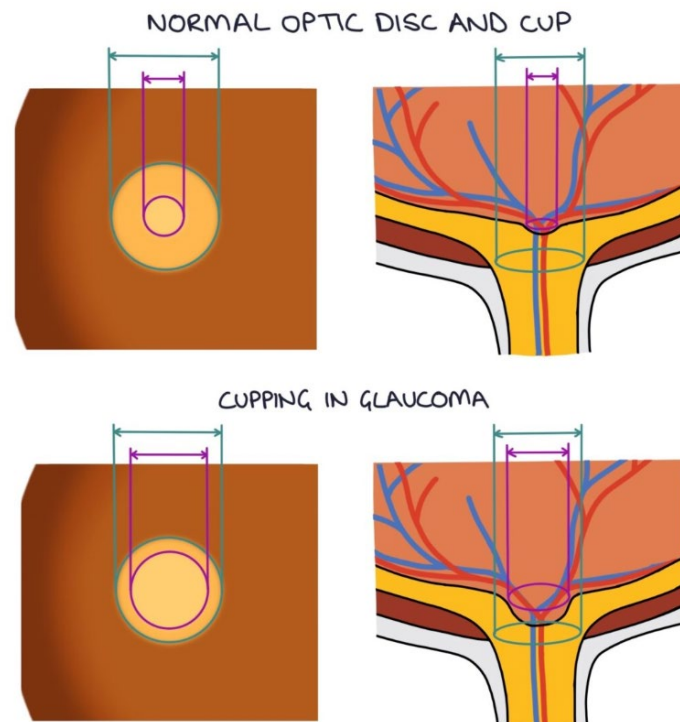


Figure 2. Cupping in Glaucoma

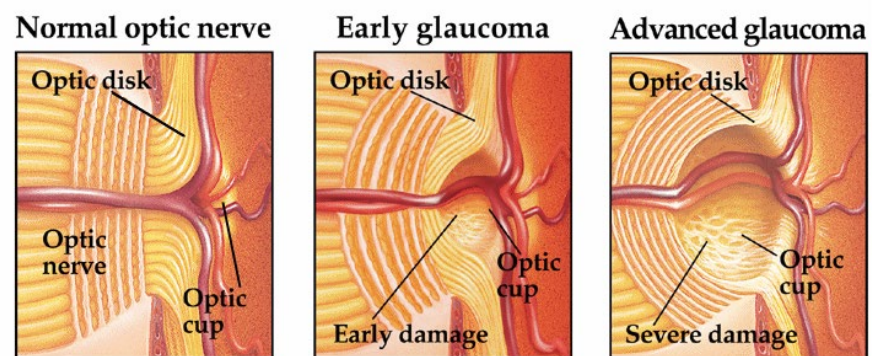


Figure 3. Eyes Optic Nerve



Figure 4. Human vision normal vs glaucoma

B. Unmet Need

Currently, the most common way to detect **Glaucoma** is using a **Dilated eye exam** with a Fundus Camera to get the fundus image. Then the clinician needs to identify the boundary contours of the cup and disc and classify them into Positive or Negative Glaucoma. Glaucoma detection is a challenging and lengthy process, relying on multiple examinations and clinical expertise. Detecting GLAUCOMA is a time-consuming and manual process that requires a trained clinician to examine and evaluate digital color fundus photographs of the retina. By the time human readers submit their reviews, often a day or two later, the delayed results lead to lost follow-up, miscommunication, and delayed treatment. Clinicians can identify GLAUCOMA by the presence of lesions associated with the vascular abnormalities caused by the disease. While this approach is effective, its resource demands are high. The expertise and equipment required are often lacking in areas where the rate of diabetes in local populations is high and GLAUCOMA detection is most needed. As the number of individuals with diabetes continues to grow, the infrastructure needed to prevent blindness due to GLAUCOMA will become even more insufficient. The need for a comprehensive and automated method of GLAUCOMA screening has long been recognized, and previous efforts have made good progress using image classification, pattern recognition, and machine learning with color fundus photography as input to classify Positive vs. Negative Glaucoma.

C. Objective

This study aims to develop Negative Fundus Glaucoma vs Positive Fundus Glaucoma using a Convolutional Neural Network (CNN).

II. Material

A. Datasets

The datasets of Fundus Glaucoma are from Kaggle.com. This website provides the datasets for Deep Learning. Datasets are provided in this link: <https://www.kaggle.com/datasets/sabari50312/fundus-pytorch>. There are more than 8000 Images of Fundus Glaucoma, from many types and sources. For more details you can see this table below:

Ref	Class	Type Data	Data	
			POSITIVE	NEGATIVE
Total Datasets			3621	5975
RefA1	Fundus Glaucoma Detection Data	BEH	127	334
		CRFO	34	18
		DRH	58	22
		EyePACS	2282	0
		FIVES	103	168
		G1020	207	502
		HRF	11	10
		JSIEC	8	32
		LES	0	11
		OIA	195	2901
		ORIGA	168	482
		PAPILLA	62	243
		REFUGE	56	517
		SJCHOI	70	210
RefB1	Glaucoma-dataset-master	-	139	225
RefB2	Retina dataset master	-	101	300

The datasets used for training are around 7170 images of Fundus Glaucoma, which contains 3440 Negative images and 3530 Positive images. Besides that, there are 200 images for testing datasets (100 Negative images and 100 positive images). Due to the bad images, and to obtain uniform datasets, sorted images are needed. The other reason is to use balanced data to train the deep learning network.



Figure 5. Datasets for training

B. Input Layer

The datasets were obtained with various sizes, ratios, and types, which means before doing the training, the datasets must be processed into the same type. The size which used is 512 x 512 pixels with a ratio of 1:1 and saved as . a PNG extension. Although ResNet18 only accepts 224*224*3 as an input, but for training is not always at that size.

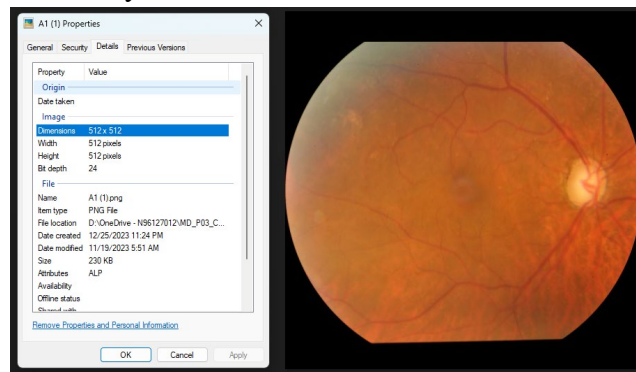


Figure 6. Data image input size

And one more thing to do before training, since these datasets are from various of type images, to get the best result for training the datasets must be shuffled into several groups, which group must contain from all types of dataset images with the same amount.

III. Method

A. Flow Chart

The flowchart of the project is as follows:

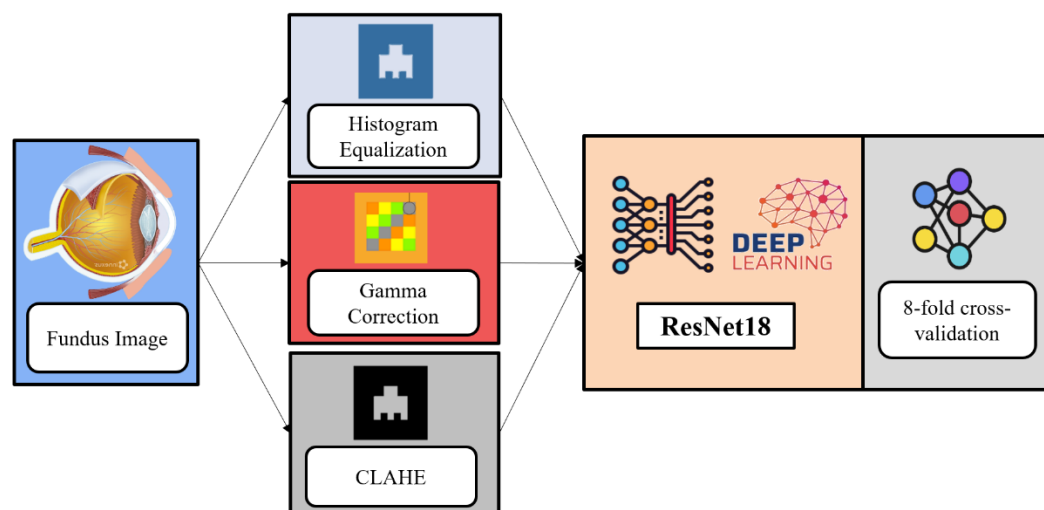


Figure 7. Project Flow Chart

B. Image Processing

The datasets of Fundus Glaucoma Images come with many different conditions. Some images are very dark, light, blur and difficult to visualize, even the image is too bad for training. Before training, Image Processing is a good choice for uniforming the dataset images. There are so many image processing techniques but in this project used three preprocessing technic, such as Histogram Equalization, Gamma Correction, and CLAHE.

1. Histogram Equalization

Histogram Equalization is a method that usually increases the global contrast of many images, especially when the usable data of the image is represented by close contrast values. Through this adjustment, the intensities can be better distributed on the histogram. This allows for areas of lower local contrast to gain a higher contrast. Histogram equalization accomplishes this by effectively spreading out the most frequent intensity values.

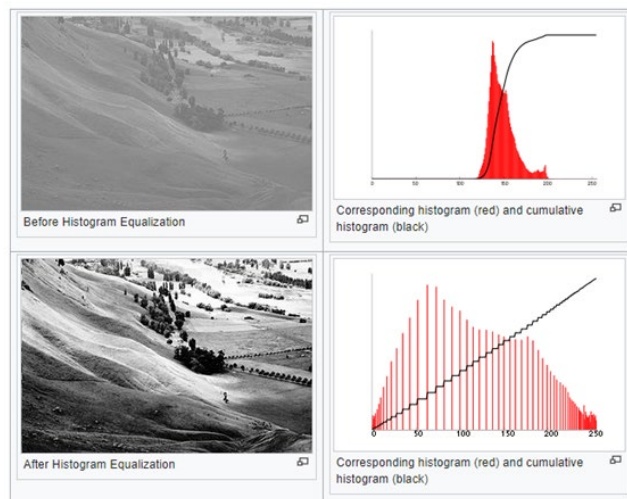


Figure 8. How Histogram Works

A key advantage of the method is that it is a fairly straightforward technique and an invertible operator. In theory, if the histogram equalization function is known, then the original histogram can be recovered. The calculation is not computationally intensive. In particular, the method can lead to better views of bone structure in x-ray images, and better detail in photographs that are over or under-exposed. A disadvantage of the method is that it is indiscriminate. It may increase the contrast of background noise while decreasing the usable signal.

We apply this image processing technique to retina images. The result shows better detail on blood vessels, improves the perception by ignoring the color perception, and works well, especially on the relatively dark picture.



Figure 9. Original vs Histogram Equalization Images

2. Gamma Correction

Gamma correction is a nonlinear adaptation applied to each and every pixel value. Generally, linear methods like addition, subtraction, and multiplication are applied to all the pixels. Gamma correction is responsible for performing nonlinear methods on the pixels of the input image and thereby remodeling the saturation of the image. (Principles of Color and Appearance Measurement, 2014). The effect of gamma correction on an image: The original image was taken to varying powers, showing that powers larger than 1 make the shadows darker, while powers smaller than 1 make dark regions lighter.

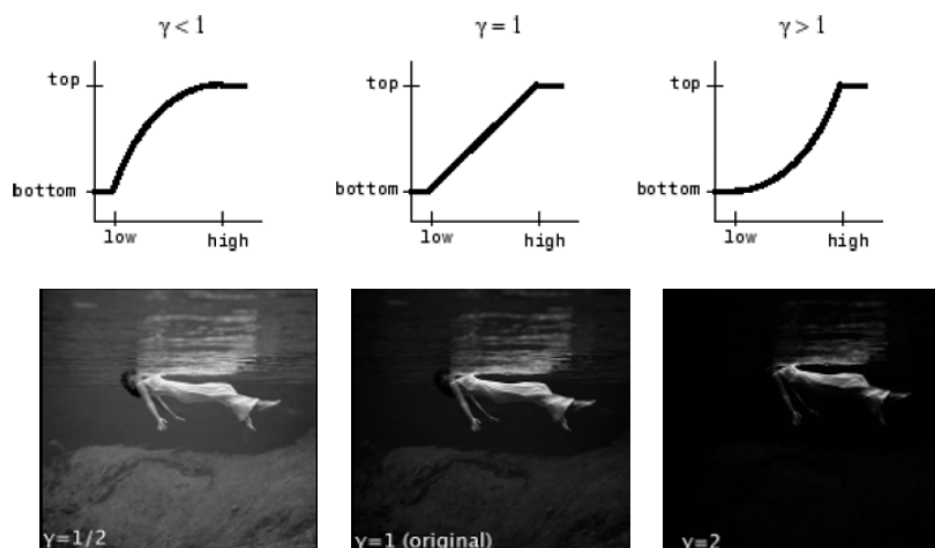


Figure 10. How does Gamma Correction work

The default gamma value for an image is one. If the image sets the value to something below one, the images will tend to be whiter, and if the image sets more than one, the images will tend to be darker. The reason gamma correction is used is because Fundus Image should have white spots we know with Optic Nerve, and make those white spots more noticeable by making the other become darker or setting the gamma into 2.



Figure 11. Original vs Gamma Correction Image

3. CLAHE

CLAHE or contrast-limited adaptive histogram equalization. This is the variant of Adaptive histogram equalization (AHE) that limits contrast amplification to reduce noise amplification. While histeq works on the entire image, CLAHE operates on small regions in the image, called tiles. The term "tiles" refers to small, distinct sections of the image that are processed independently. When applying CLAHE, an image is divided into smaller regions or "tiles" for the purpose of performing localized histogram equalization.

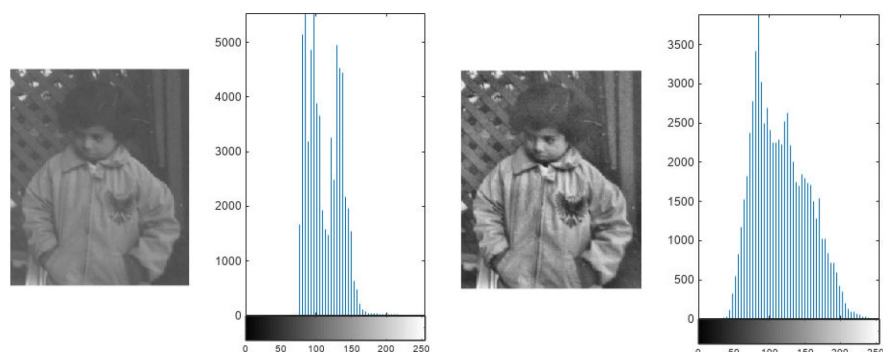


Figure 12. How CLAHE works

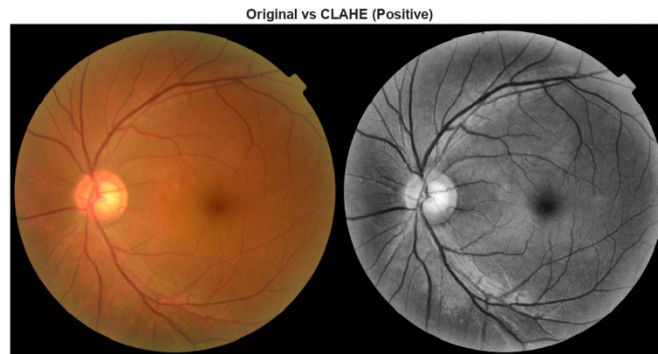


Figure 13. Original vs CLAHE image

C. Deep Learning

A Convolutional Neural Network (ConvNet/CNN) is a Deep Learning algorithm which can take in an input image, assign importance (learnable weights and biases) to various aspects/objects in the image, and be able to differentiate one from the other. (Saha,2018). In this project used ResNet18 to train a model for Fundus Glaucoma Images Datasets.

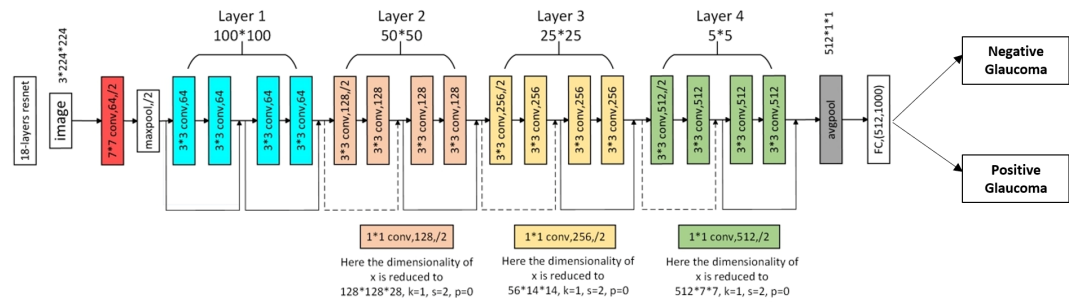


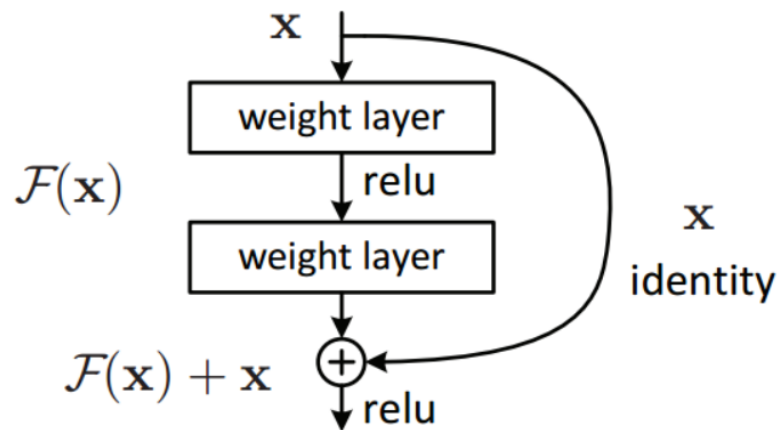
Figure 14. ResNet18 Architecture

ResNet-18 is known as Residual Network. The number 18 denotes the number of layers and is computed as follows: $1 + (4+4+4+4) + 1$. Its architecture comprises what we called Residual Blocks. The basic building blocks for ResNets are the conv and identity blocks. The core idea of ResNet is introducing a so-called “identity shortcut connection” that skips one or more layers. For a deep neural network, it is not just simply to add more layers. The problem appears: “degradation” or “vanishing gradient”. Imagine that at each block of the convolutional layer you downsize the feature maps to some point your feature will become too small and it can not be downsized anymore. Keeping downsizing just increases the error and no further learning can be made.

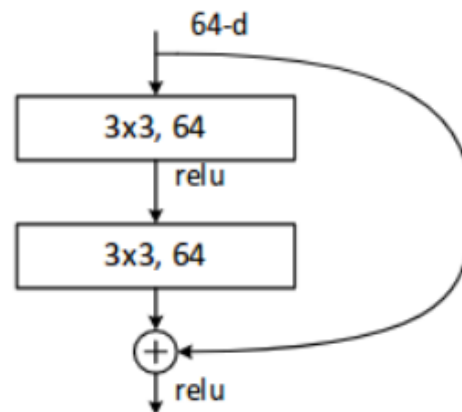
Table 1. ResNet-18 Architecture.

Layer Name	Output Size	ResNet-18
conv1	$112 \times 112 \times 64$	$7 \times 7, 64$, stride 2
conv2_x	$56 \times 56 \times 64$	3×3 max pool, stride 2 $\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$
conv3_x	$28 \times 28 \times 128$	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$
conv4_x	$14 \times 14 \times 256$	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$
conv5_x	$7 \times 7 \times 512$	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$
average pool	$1 \times 1 \times 512$	7×7 average pool
fully connected	1000	512×1000 fully connections
softmax	1000	

In a network with residual blocks, each layer feeds into the next layer and directly into the layers about 2–3 hops away.



Resnet-18 contains 18 main layers, each ResNet block is 2 layers deep (except for conv1, as the figure above)



Some of the convolutional layers are followed by max-pooling layers, Batch Normalization, and ReLu. At the end, an Average Pooling layer is connected with a fully connected layer. SoftMax and Classification are the final layers of other neural networks.

D. Validation Method

K-fold cross-validation is used in this project, especially 10-fold cross-validation. The data set is split into a K number of sections/folds where each fold is used as a testing set at some point.

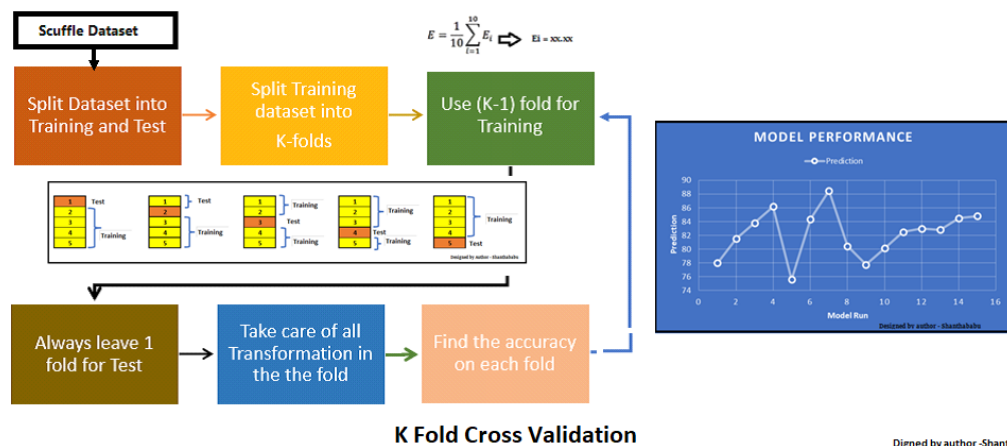


Figure 15. K-Fold Cross Validation

Cross-validation is a resampling procedure used to evaluate machine learning models on a limited data sample. It is a popular method because it is simple to understand and generally it results in a less biased or less optimistic estimate of the model skill than other methods, such as a simple train/test split.

A higher k (number of folds) means that each model is trained on a larger training set and tested on a smaller test fold. In theory, this should lead to a

lower prediction error as the models see more of the available data. A lower k means that the model is trained on a smaller training set and tested on a larger test fold. Here, the potential for the data distribution in the test fold to differ from the training set is bigger, and we should thus expect a higher prediction error on average.

E. Evaluation Parameters

A multi-categorical classification model can be evaluated by the sensitivity and specificity of each possible class. A model that is great for predicting one category can be terrible for the others. These evaluation metrics allow us to analyze models for different purposes. You have to decide which evaluation metric is best for your project. The output of the Confusion matrix is Accuracy, Accuracy : measures the total number of correct classifications divided by the total number of cases. Besides that, there is another evaluation parameter: That is: Sensitivity is the metric that evaluates a model's ability to predict true positives of each available category. Specificity is the metric that evaluates a model's ability to predict the true negatives of each available category. Specificity determines a model's ability to predict if an observation does not belong to a specific category.

Confusion Matrix		Actual Class		$\text{Sensitivity} = \frac{TP}{TP + FN}$	$\text{Specificity} = \frac{TN}{TN + FP}$
		A	B		
Predicted Class	A	TP	FP		
	B	FN	TN		
Total		P	N	$\text{Accuracy} = \frac{TP + TN}{P + N}$	

Note:

A = Condition Positive set
B = Condition Negative set

TP = True Positive : FP = False Positive
FN = False Negative : TN = True Negative

Figure 16. Evaluation Parameters Formula

IV. Result

A. Image Processing Result

This is the result of the image processing. in the HE images, the color is more distributed, the black looks darker and the white color looks whiter. As for gamma correction, because the gamma is set to 2, it means that the image would be darker, which makes the image darker and appears/highlights the Optic Nerve. And for the CLAHE as you can see, the image highlights the color becoming gray but minimizes the noise. And mention the retinal vessels more clearly.

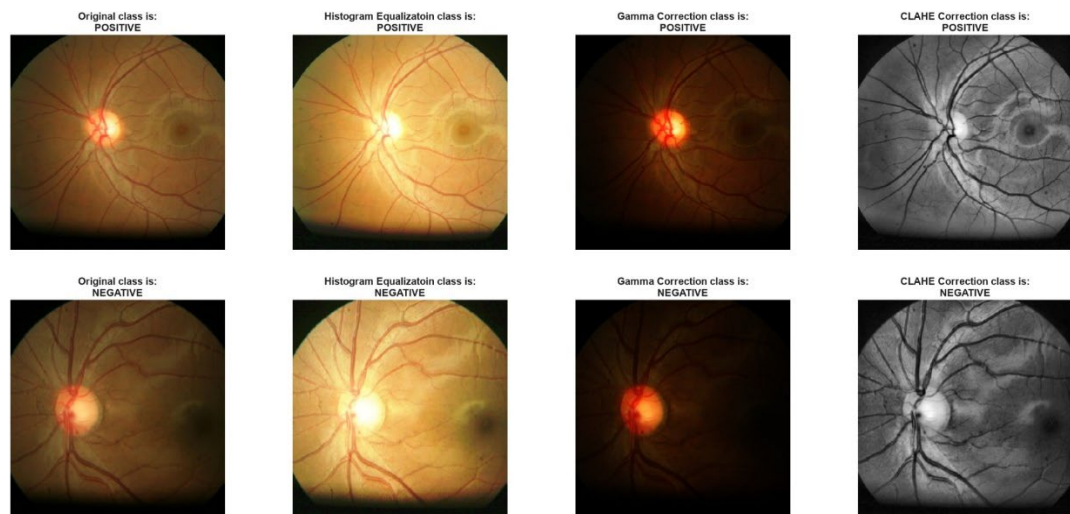


Figure 17. Image Processing Result

B. Deep Learning Performance

After training the datasets, including with the 3 IM method, In this project the accuracy, sensitivity, and specificity were more than 90 percent. And the best Model accuracy is by the original datasets and Gamma Correction datasets which get 91 percent. That is shown in the table below:

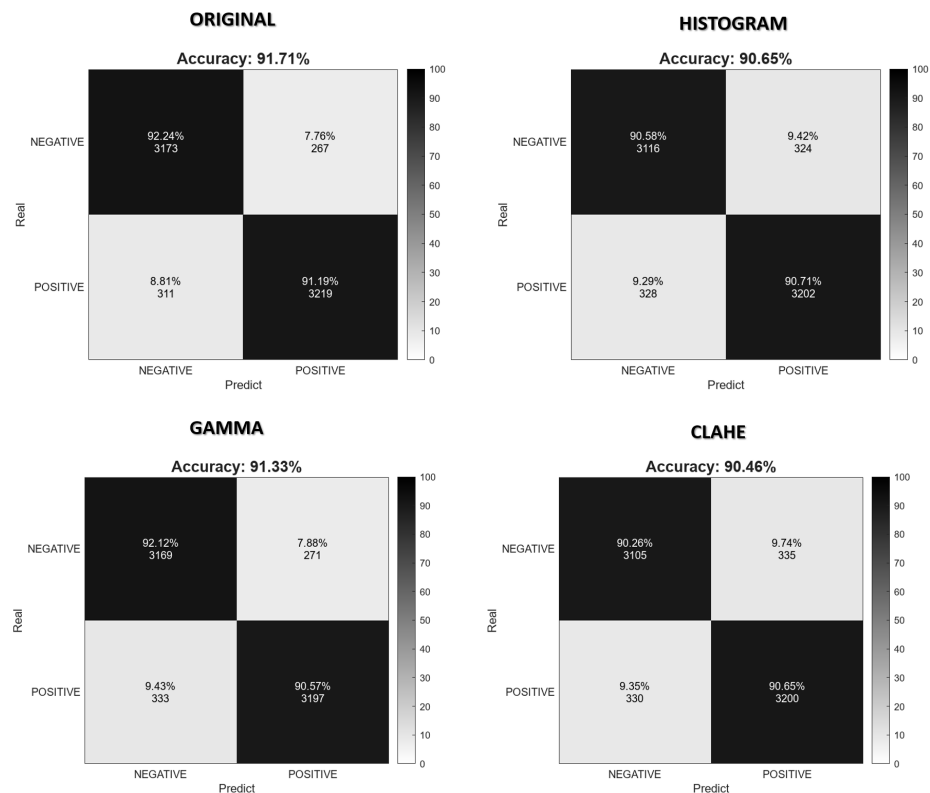
Evaluation Parameter	Image Types			
	Original	Histogram Equalization	Gamma Corrections	CLAHE
Accuracy (%)	91.707	90.645	91.334	90.469
Sensitivity (%)	91.714	90.644	91.344	90.466
Specificity (%)	91.714	90.644	91.344	90.466

V. Discussion

This project proposes the use of Fundus Glaucoma Images as input to a CNN for classifying Negative and Positive Glaucoma. With an accuracy of 91.71% with the original image, the ResNet18 model has achieved high performance. The proposed method is accurate with the literature we refer to as image enhancement and able to increase the accuracy of the deep learning network.

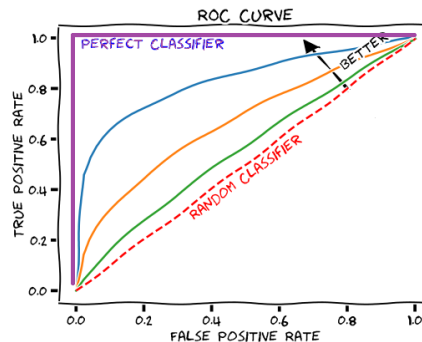
A. Discussion on Confusion Matrix

A confusion matrix, also known as an error matrix, is a special table structure that permits visualization of the performance of an algorithm, often a supervised learning one, in the field of machine learning and specifically the problem of statistical application (in unsupervised learning it is usually called a matching matrix). The instances in a predicted class are represented in each row of the matrix, while the occurrences in an actual class are represented in each column (or vice versa). As a result of our findings, we can estimate that the average accuracy of the four types of images is 91.04 percent.



B. Discussion on Receiver Operating Characteristics (ROC)

1. ROC Interpretation



Another way to interpret the performance of the model is by observing the ROC. The ROC is made of False Positive Rate (FPR) as the x-axis and True Positive Rate as (TPR) as the y-axis. The curve on ROC is made of points that are plotted for every possible threshold value between 0.0 and 1.0.

$$TPR = \frac{TP}{TP + FN}; FPR = \frac{FP}{FP + TN}$$

The ROC curve is great for choosing a threshold. Its shape contains a lot of information:

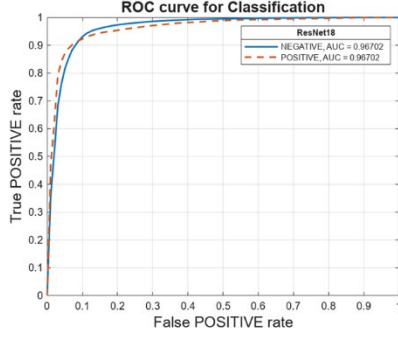
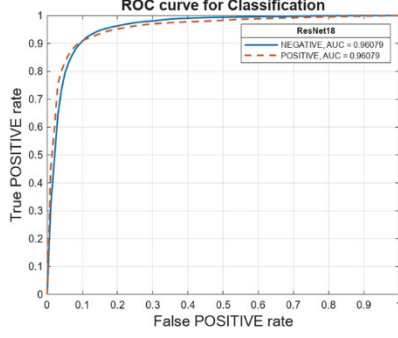
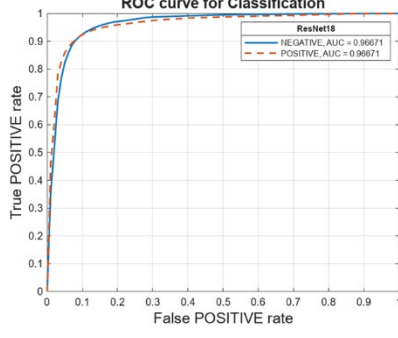
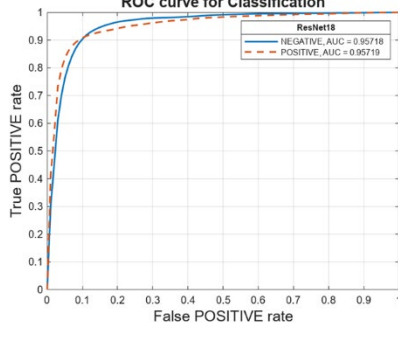
- Less false positives and more true negatives are shown by lower values on the x-axis of the plot.
- Greater true positives and smaller false negatives are shown by higher values on the y-axis of the plot.
- A good model has high y values at low x values.

The ROC curve is very useful for a few additional reasons:

- The curve of different models can be compared directly in general or for different thresholds.
- The area under the curve (AUC) can be used as a summary of the model performance.

2. ROC Models

Datasets of Fundus Glaucoma Images

No.	Description	Images
1.	Original: AUC = 0.96702	
2.	Histogram Equalization: AUC = 0.96079	
3.	Gamma Correction: AUC = 0.96671	
4.	CLAHE: AUC = 0.95719	

C. Discussion on Score-CAM

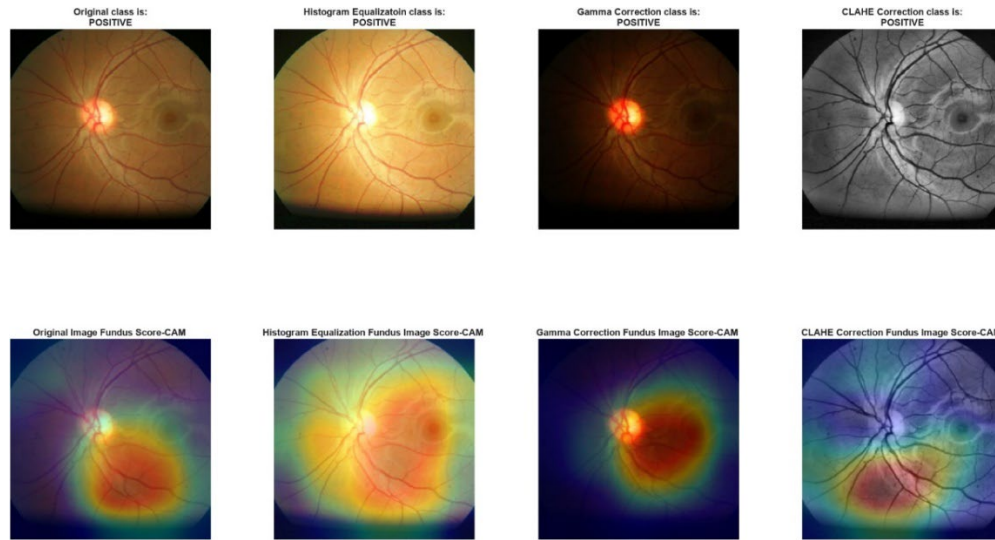


Figure 18. Score-CAM visualization of Fundus Glaucoma images

Score-CAM, or Score-Weighted Visual Explanations, is a technique for determining where the network is picking up information from the Fundus Glaucoma Images. The network can train to classify pictures based on relevant and non-related regions, which may be confirmed using Score-CAM-based heat maps created by the data.

As you can see from the image, the decision-making by the CNN models in the original Fundus Glaucoma image does not always come from the Optic Nerve. The areas of the Fundus Glaucoma images that are mostly contributing to CNN's decisions are not always the optic when plain Fundus Glaucoma images are used for classification. The conclusion is a network could be learned from an irrelevant area, therefore a method of validation like Score-CAM-based is very important to know the ratatability of the trained network. The conclusion is that a network may be learned from a completely unrelated field. As a result, a validation approach such as Score-CAM-based is critical for determining the trained network's dependability.

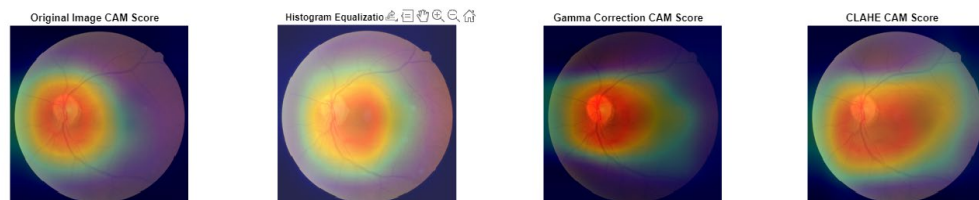
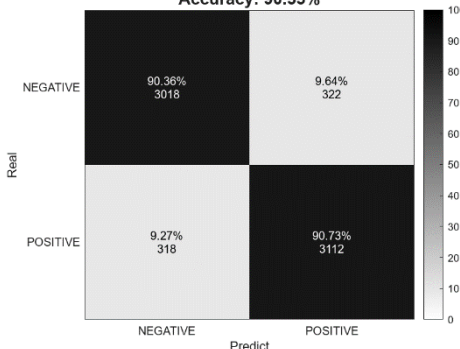
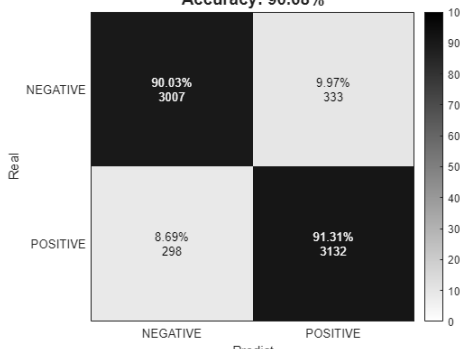
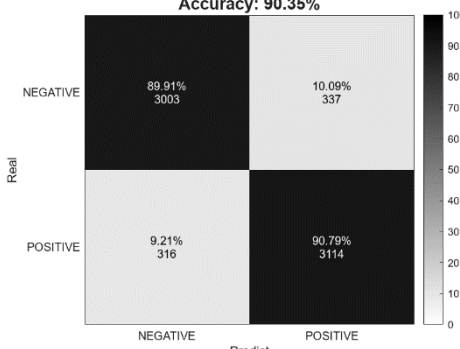


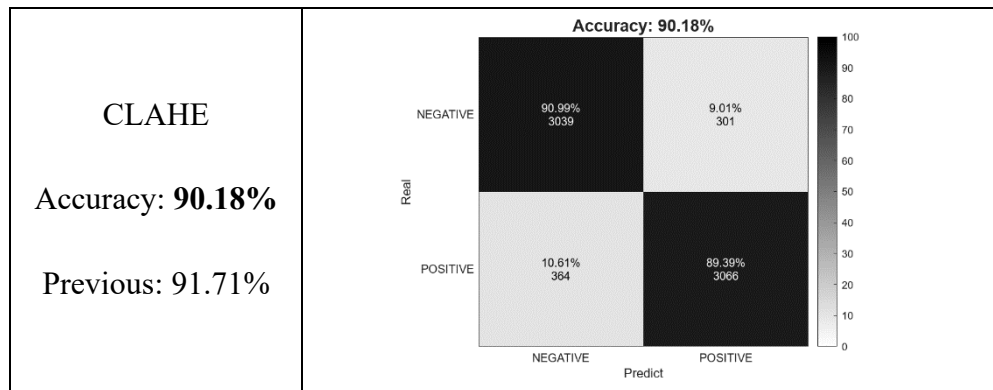
Figure 19. Correctly Classified Glaucoma Score-CAM

D. Discussion on Accuracy without Test Datasets

To answer the question asked, What is the accuracy when working on the 200 images never used in training?

After rerunning the Training without test datasets, which are 3340 Negative Images and 3430 Positive Images, the accuracy I obtained is around 90%.

Original Datasets Accuracy: 90.55% Previous: 91.71%	Accuracy: 90.55%  Real NEGATIVE POSITIVE Predict NEGATIVE POSITIVE
Histogram Equalization Accuracy: 90.68% Previous: 90.65%	Accuracy: 90.68%  Real NEGATIVE POSITIVE Predict NEGATIVE POSITIVE
Gamma Correction Accuracy: 90.35% Previous: 90.46%	Accuracy: 90.35%  Real NEGATIVE POSITIVE Predict NEGATIVE POSITIVE



In this training, the best accuracy is not Original Datasets, but the best accuracy is Histogram Equalization Datasets with Accuracy 90.68%. Almost in all datasets, the accuracy is decreasing, even though not too significantly. However, in Histogram Equalization the accuracy increases, from 90.65% to 90.68%, and becomes higher from the others.

VI. Conclusion

Convolutional Neural Network based on color fundus image can classify Glaucoma with the best accuracy of 91.71% on Original Training Images. Here is the GUI of the app, and the link for the demo App: https://youtu.be/GEPJ71UQ_gg

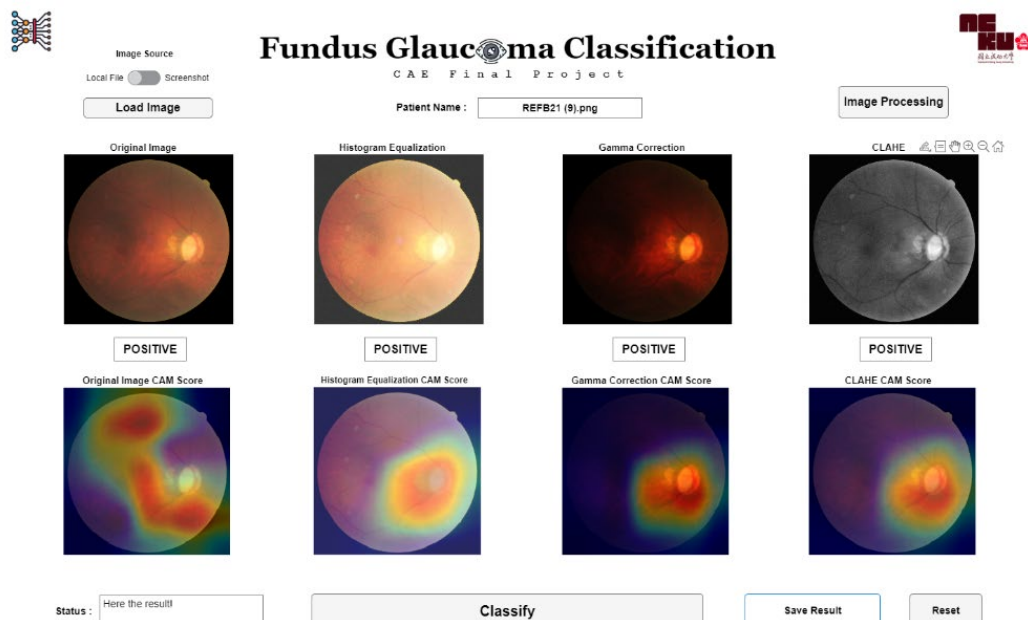


Figure 20. GUI of Fundus Glaucoma Classification App

VII. Reference

- Saba T, Bokhari STF, Sharif M, Yasmin M, Raza M. Fundus image classification methods for the detection of glaucoma: A review. *Microsc Res Tech.* 2018; 81:1105–1121. <https://doi.org/10.1002/jemt.23094>
- Hemelings, R., Elen, B., Barbosa-Breda, J. *et al.* Deep learning on fundus images detects glaucoma beyond the optic disc. *Sci Rep* **11**, 20313 (2021). <https://doi.org/10.1038/s41598-021-99605-1>
- Lauren J. Coan, Bryan M. Williams, Venkatesh Krishna Adithya, Swati Upadhyaya, Ala Alkafri, Silvester Czanner, Rengaraj Venkatesh, Colin E. Willoughby, Srinivasan Kavitha, Gabriela Czanner. Automatic detection of glaucoma via fundus imaging and artificial intelligence: A review, 2023, Pages 17-41, <https://doi.org/10.1016/j.survophthal.2022.08.005>

VIII. Individual Contribution to this Project

I don't have any members, but these are what I did for this project.

1. Find The Original Image Dataset
2. Choose The Previous Research to be Referred and Compared
3. Image Processing
4. Training the datasets
5. Verify The Result
6. Evaluate The Performance of The Model

IX. Feedback on this Project and Class

Studying this course and working on the final project was something new, considering my background as a mechanical design engineer made me interested in this course when I saw the course title, although I don't know why it was a little different from what I imagined. But in the end, I liked this course, I learned new things and learned what I love and it was fun. With a very good professor, it made me enjoy the course. Even though I didn't have a member for my final project, I don't regret doing it alone, because that way I tried various things until I succeeded. Although there are many obstacles, confusion, and ignorance, if you keep trying, you will eventually succeed.